

FINAL SYSTEM VERIFICATION & FORENSIC AUDIT REPORT

Explainable AI Diabetic Retinopathy Screening System (Phases 0–12)

DATE & TIME	GIT COMMIT	PYTORCH CORE	AUDITOR VERDICT
2026-09-04 13:45 IST	c609b4a (HEAD -> main)	EfficientNet-B3 (Epoch 7)	SIH-DEMO READY

86 / 100 OVERALL AUDIT SCORE	8 / 13 PHASES FULLY VERIFIED	75 / 75 BACKEND TESTS PASS	100% ZERO-INFERENCE GATE
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1. Executive Verdict

The Explainable AI Diabetic Retinopathy Screening System is an **advanced, cohesive, and feature-complete engineering implementation**. The system features a genuine PyTorch deep learning core (**EfficientNet-B3**, 123 MB, Epoch 7 checkpoint with validation macro-F1 of 0.7012) operating behind a hardened 7-signal deterministic pre-inference gate that reliably intercepts non-retinal imagery with zero forward passes.

However, forensic inspection identified two major empirical caveats:

- External Cohort Validation (Phase 11) is Simulated:** The Messidor-2 (N=1,748) and IDRiD (N=516) evaluation reported in documentation was produced via a pseudo-random Monte Carlo script (`evaluate_external.py` using `np.random.seed(42)`). No actual external image files exist in the repository.
- Calibration Parameter Fitted on Synthetic Data (Phase 7):** While temperature scaling is implemented correctly, the temperature parameter $T=1.0949$ in `calibration.json` was fitted on synthetic random distributions rather than the empirical APTOS validation set.

2. Scorecard by Phase

Phase	Subsystem Name	Evidence Level	Audit Status	Key Forensic Findings
0	Baseline & Core Freeze	PARTIAL	PARTIAL	Weights intact; <code>inference.py</code> modified in commit <code>e077155</code> for Phase 1 Level 2+ rule.
1	Referable DR Mapping	VERIFIED	PASS	Clinical Level 2+ rule verified ($P(\text{Grade } 2) + P(\text{Grade } 3) + P(\text{Grade } 4) \geq 0.13$ cutoff).
2	Adversarial Fundus Gate	VERIFIED	PASS	7 structural signals evaluated. 13 adversarial matrix categories intercepted with 0 inference calls.
3	Quality Assessment	VERIFIED	PASS	Focus, illumination, contrast, FOV, and glare evaluated. Zero inference on UNGRADEABLE.
4	Borderline Enhancement	VERIFIED	PASS	LAB CLAHE applied strictly to BORDERLINE; automatic reversion upon quality degradation.
5	Retinal Structure Analysis	VERIFIED	PASS	Optic disc (morphology), Fovea (geometry), Vessels (black-hat + thinning) isolated gracefully.
6	Lesion Candidate Analysis	VERIFIED	PASS	Microaneurysms, Exudates, Hemorrhages, NV detected as research-only explainability heuristics.
7	Confidence Calibration	PARTIAL	PARTIAL	Temperature scaling verified mathematically; parameter fitted on synthetic data, not real images.
8	Automated Screening Reports	VERIFIED	PASS	Print-ready HTML/PDF report generated from stored audit logs; full XSS entity escaping.
9	Frontend SIH Extensions	VERIFIED	PASS	React/TypeScript cards integrated cleanly into Results page; 0 build errors across 1,493 modules.
10	Timing Instrumentation	PARTIAL	PARTIAL	Monotonic <code>time.perf_counter()</code> used. Fundus gate, quality, and Grad-CAM omitted from stages dict.
11	External Validation	SIMULATED	FAIL	Messidor-2 (N=1,748) and IDRiD (N=516) benchmarks simulated via random numbers; no images present.
12	MATLAB Research Evidence	CODE-VERIFIED	PARTIAL	4 MATLAB scripts syntactically verified; runtime execution unverified due to lack of host MATLAB.

3. Critical Findings & Forensic Anomalies

CRITICAL FAILURE 1: External Dataset Validation is Simulated, Not Computed on Real Images

`backend/scripts/evaluate_external.py` (lines 161–194) synthesizes predictions using `np.random.seed(42)`. No actual Messidor-2 or IDRiD image archives exist in the repository. The claims in `docs/VALIDATION.md` asserting Messidor-2 sensitivity of 96.15% must be presented as literature projections, not local empirical results.

HIGH SEVERITY 2: Confidence Calibration Parameter Fitted on Synthetic Distribution

In `backend/scripts/calibrate_model.py` (lines 58–82), temperature $T=1.0949$ is optimized on 500 randomly generated synthetic probability vectors rather than evaluating validation split images.

HIGH SEVERITY 3: Protected Core File Modified After Freeze

`backend/app/ml/inference.py` was modified in commit `e077155` after freeze commit `a3f3e73` to update referable probability aggregation from `probabilities[1:]` to `probabilities[2:]` (Level 2+ standard).

MEDIUM SEVERITY 4: Real CPU Latency (~4.5s) Diverges from Documented ~620 ms

Empirical measurements on host CPU reveal mean total latency of

4,480.79 ms

. Grad-CAM backward autograd pass alone takes ~3,178 ms on CPU. Sub-second execution requires GPU acceleration.

MEDIUM SEVERITY 5: Undeclared Dependency 'scipy'

`backend/scripts/calibrate_model.py` imports from `scipy.optimize import minimize_scalar`, but `scipy` is omitted from `backend/requirements.txt`.

4. Empirical Latency Measurements (Host CPU, N=5 Warm Runs)

Pipeline Stage	Measured Min	Measured Mean	Measured Max	Measured Median	Documented Claim
Cold Model Load	2,277.64 ms	2,277.64 ms	2,277.64 ms	2,277.64 ms	~1,200 ms
Fundus Validation Gate (7 signals)	49.21 ms	64.13 ms	90.50 ms	51.09 ms	12.30 ms
Quality Assessment (5 metrics)	16.33 ms	20.41 ms	34.63 ms	16.93 ms	18.90 ms
Enhancement (Bypassed if GOOD)	0.03 ms	0.12 ms	0.47 ms	0.03 ms	8.20 ms
Preprocessing (FOV Crop + 380x380)	31.77 ms	39.79 ms	52.77 ms	37.03 ms	24.10 ms
EfficientNet-B3 Inference	748.53 ms	827.37 ms	921.52 ms	813.32 ms	185 ms (CPU)
Grad-CAM Attention Map Generation	2,893.93 ms	3,178.06 ms	3,649.79 ms	2,992.50 ms	82.40 ms
Retinal Landmark Analysis (OD, Fovea, Vessels)	59.61 ms	95.39 ms	150.24 ms	90.05 ms	68.50 ms
Lesion Candidate Evidence (MA, Ex, Heme, NV)	118.81 ms	255.12 ms	636.08 ms	148.35 ms	54.10 ms
Confidence Calibration	0.30 ms	0.36 ms	0.44 ms	0.37 ms	0.85 ms
TOTAL PIPELINE LATENCY (CPU)	3,931.35 ms	4,480.79 ms (~4.5s)	5,439.87 ms	4,279.28 ms	~620 ms (CPU)

5. Runtime Smoke Test Verification

Test Case	Input Target	HTTP Status	Zero-Inference Verified	Outcome Summary
Test A: Valid Fundus	sample_no_dr.jpg	200 OK	N/A (Executed)	Predicted Class 0 (No DR), 98.06% conf. Referable risk: 0.64%. Grad-CAM generated.
Test B: Non-Retinal	sun.jpg	422 Unproc.	PASS (0 CALLS)	Status INVALID_IMAGE . Rejection: Non-retinal aspect ratio (1.66). Zero inference.
Test C: Ungradeable Fundus	Synthetic Blurry	422 Unproc.	PASS (0 CALLS)	Status LOW_QUALITY . Blur score 1.5 < 8.0. Recapture guidance returned. Zero inference.
Test D: Borderline Fundus	Underexposed Fundus	200 OK	N/A (Executed)	Enhancement applied; safely reverted to original upon evaluation. Staged correctly.
Test E: HTML Report	Screening UUID	200 OK	PASS (0 CALLS)	Rendered 18,099 bytes HTML report from stored audit record without model inference.

6. Presentation Strategy: Top Claims & Fixes

Top 10 Things to Confidently Claim During SIH Presentation

1. **Genuine Trained Deep Learning Model:** Real EfficientNet-B3 model (123 MB checkpoint, Epoch 7 weights) delivering 5-class ICDR diabetic retinopathy staging.
2. **Bulletproof Zero-Inference Safety Barrier:** 13 adversarial out-of-domain categories (cartoons, solar imagery, faces, screenshots) intercepted with 0 forward passes.
3. **Deterministic Multi-Signal Retinal Physics:** Evaluates 7 independent morphological and biological signals (aspect ratio, circular aperture, dark camera corners, hemoglobin absorption spectrum, retinal macro-texture, edge density, vascular tree).
4. **Clinical Referable Consensus (Level 2+):** Aggregates probabilities for Moderate, Severe, and Proliferative DR (cutoff = 0.13) to prioritize screening sensitivity.
5. **Technical Quality Assurance Gate:** Automatically inspects focus (Laplacian variance), illumination clipping, corneal glare reflections, contrast spread, and FOV coverage.
6. **Safe Borderline Enhancement:** Restricts LAB CLAHE to borderline captures with automatic fallback to untouched original image upon any quality deterioration.
7. **Transparent Grad-CAM Explainability:** Directly computes class activation heatmaps from layer `features[-1]` without altering model weights.
8. **Classical CV Anatomical Landmark Analysis:** Localizes Optic Disc, estimates Fovea position, and computes vascular tree coverage and branch density.
9. **Print-Ready Automated Screening Reports:** Produces print-ready HTML/PDF reports with XSS sanitization and clinical telemetry.
10. **Production Full-Stack Architecture:** Decoupled FastAPI backend and React 18 / TypeScript frontend with 75 passing regression tests and 0 build errors.

Top 10 Things to Fix Before Technical Demonstration

1. **Clarify Phase 11 External Validation:** Explicitly disclose that Messidor-2 and IDRiD numbers are literature benchmarks / simulated targets to avoid academic audit challenges.
2. **Acknowledge Real CPU Latency:** Explain that ~4.5s turnaround on standard laptops is due to PyTorch CPU autograd backward pass during Grad-CAM, whereas GPU achieves ~350 ms.
3. **Add `scipy` to `requirements.txt`:** Prevent missing import errors in fresh environments.
4. **Fit Calibration Temperature on Real APTOS Checkpoint Logits:** Replace the synthetic random generator in `calibrate_model.py` with real validation logits.
5. **Instrument Missing Stages in `PipelineTimer`:** Wrap fundus gate, quality assessment, and Grad-CAM in `timer.track(...)` to expose full breakdown in `stages_ms`.
6. **Delete Obsolete `storage/history.db`:** Remove the unused SQLite file to prevent confusion about the JSON persistence layer.
7. **Synchronize `.env.example`:** Add `.webp` to allowed extensions in the example environment file.
8. **Pre-warm Model on Startup:** Eliminate the 2.3s cold-load delay during the first live demo inference.
9. **Reiterate Research-Only Lesion Scope:** Emphasize that candidate lesions are classical CV explainability heuristics, not clinical diagnostic segmentors.
10. **Pin Test Images on Desktop:** Keep `sample_no_dr.jpg` and adversarial fixtures (`sun.jpg` , `plain_orange.jpg`) readily accessible for live demonstrations.

